Foundations of Arithmetic Power Geometry XII: Conductor–Modular Degree Dominance and Boundary-Control Growth Estimates

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Abstract

Arithmetic Power Geometry (APG) studies algebraic closure relations under exponent deformation and their analytic realization on compactified modular curves. Earlier volumes established entropy-governed local closure defects, stabilized information geometry, integrated defect functionals, scale correction, discrete arithmetic regularization, conditional APG–Arakelov height coupling, discrepancy–source compatibility, the Spectral–Green Bridge, source concentration, entropy-controlled spectral energy, and spectral-gap stability for a canonical bounded Green-renormalized APG metric. This paper develops the twelfth volume of the APG program. Its purpose is to connect APG analytic size to conductor and modular-degree growth before the final spectral-conductor synthesis of APG XIII.

Let $X = X_0(N)^*(\mathbb{C})$ and let $\pi : X_0(N)^* \to E$ be the modular parametrization attached to the APG–Arakelov setting. We introduce the canonical APG boundary package

$$\mathcal{B}_{\mathrm{APG}}(N, \pi) = B_{\mathrm{cusp}} + B_{\mathrm{vert}} + B_{\mathrm{Green}},$$

where the three terms measure cusp-width complexity, vertical arithmetic complexity, and Green-energy metric distortion. The main unconditional result of this manuscript is the conductor-scale upper dominance theorem

$$B_{\mathrm{cusp}} + B_{\mathrm{vert}} + B_{\mathrm{Green}} \leq C \deg(\pi) \log N,$$

under the canonical normalization inherited from APG VI–XI. We also prove a direct non-circular entropy-to-boundary lower comparison in the logarithmic case $A = 1$, and we formulate the general Spectral Entropy Boundary Realization Problem for $A \geq 1$. If this realization property holds, then APG XII yields the desired conductor–modular degree dominance estimate

$$H(W)^2 (\log N)^A \leq C_3 \deg(\pi) \log N.$$

Consequently, APG XII isolates the precise final arithmetic-growth input required by APG XIII. The paper does not claim a proof of Fermat's Last Theorem, the abc conjecture, the Szpiro conjecture, the Birch and Swinnerton-Dyer conjecture, or the full APG–Arakelov

Projection Theorem. Its contribution is a rigorous boundary-growth theorem and a precise reduction of the remaining APG spectral-conductor obstruction to an entropy-to-boundary realization theorem.

Keywords: Arithmetic Power Geometry; modular degree; conductor growth; Arakelov geometry; spectral gap; entropy; modular curves; Green functions; boundary control; Laplacian; cusp widths; vertical components.

1 Introduction

Arithmetic Power Geometry (APG) is a deformation-theoretic framework for measuring how algebraic closure relations change when exponents are treated as deformation parameters rather than as fixed integers. The first volume introduced the Power-Deformation Space and proved that the leading local closure defect near the Euclidean target is governed by Shannon entropy (Akhtar, 2026a; Shannon, 1948; Cover and Thomas, 2006). APG II developed information-geometric stabilization and coordinate-stability principles in the language of information geometry (Akhtar, 2026b; Amari, 2016; Nielsen, 2020). APG III introduced the integrated closure-defect functional, while APG IV and APG V added scale correction, coordinate-dependent damping, and discrete prime-chain regularization (Akhtar, 2026c,d,e).

The later APG program moves toward the APG–Arakelov architecture. APG VI formulated a conditional APG–Arakelov height-coupling program over compactified modular curves and Frey-type elliptic curves (Akhtar, 2026f; Frey, 1986; Faltings, 1983, 1984). APG VII identified the discrepancy–source compatibility mechanism and reduced the missing projection step to three boundary profiles: cusp-width control, vertical component control, and Green-energy metric distortion (Akhtar, 2026g; Arakelov, 1974; Soulé, 1992; Zhang, 1993). APG VIII then proved a Spectral–Green Bridge, APG IX proved source concentration and cusp-integrable density bounds, APG X derived spectral entropy control, and APG XI established spectral-gap stability for a canonical bounded Green-renormalized APG metric (Akhtar, 2026h,i,j,k).

The present manuscript develops APG XII. The target of the broader APG IX–XIII route is the spectral-conductor inequality

$$\lambda_1(\Delta_{\mathrm{APG}})^{-1} \left\| \rho_{\mathrm{APG}} - \frac{1}{\mathrm{Vol}(X)} \right\|^2_{L^2(X)} \leq C \deg(\pi) \log N. \tag{1}$$

APG IX supplies the source estimate

$$\left\| \rho_{\mathrm{APG}} - \frac{1}{\mathrm{Vol}(X)} \right\|^2_{L^2(X)} \leq C_1 H(W)^2, \tag{2}$$

where $H(W)$ is the Shannon entropy of the APG weight distribution. APG XI supplies the canonical spectral-gap estimate

$$\lambda_1(\Delta_{\mathrm{APG}}) \geq \frac{c}{(\log N)^A}, \tag{3}$$

for a fixed exponent $A \geq 1$. Combining (2) and (3) reduces (1) to the conductor-growth estimate

$$H(W)^2(\log N)^A \leq C_3 \deg(\pi) \log N. \tag{4}$$

Equation (4) is the APG XII milestone.

This paper makes two claims and one reduction. First, it proves that the canonical APG boundary package is conductor-dominated:

$$B_{\text{cusp}} + B_{\text{vert}} + B_{\text{Green}} \leq C \deg(\pi) \log N. \tag{5}$$

Second, it proves a direct non-circular entropy-to-boundary lower comparison when the spectral exponent is $A = 1$. Third, for general $A \geq 1$, it isolates the exact missing statement as the Spectral Entropy Boundary Realization Problem. This is deliberately not hidden as an assumption inside the proof. It is stated as the final arithmetic-growth problem needed by APG XIII.

The manuscript uses standard background from modular curves, Arakelov geometry, Diophantine geometry, and spectral theory (Shimura, 1971; Deligne, 1971; Diamond and Shurman, 2005; Hartshorne, 1977; Bombieri and Gubler, 2006; Hindry and Silverman, 2000; Lang, 1983; Silverman, 1986, 2007; Iwaniec, 2002; Selberg, 1965). Its arithmetic setting is also informed by classical number theory, modular curves, deformation rings, level lowering, modularity lifting, and Diophantine approximation (Borevich and Shafarevich, 1966; Borel, 1997; Katz, 1973; Mazur, 1977; Diamond, 1996; Ribet, 1990; Serre, 1987; Taylor and Wiles, 1995; Wiles, 1995; Vojta, 1987).

2 Scope, Notation, and Logical Status

Let

$$X = X_0(N)^*(\mathbb{C}) \tag{6}$$

be the compactified complex modular curve of level N, and let

$$\pi : X_0(N)^* \longrightarrow E \tag{7}$$

be the modular parametrization into an elliptic curve E associated with the APG–Arakelov setting. The notation $\deg(\pi)$ denotes the modular degree. The APG metric is fixed as

$$g_{\text{APG}} = e^{2\phi_{\text{APG}}} g_{\text{std}}, \tag{8}$$

where g_{std} is a compact Arakelov-compatible reference metric and ϕ_{APG} is the bounded Green-renormalized conformal factor from APG XI.

The first positive eigenvalue of the positive APG Laplacian is denoted by $\lambda_1(\Delta_{\text{APG}})$. The APG source density is ρ_{APG}, and its mean-zero fluctuation is

$$F_{\text{APG}} = \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)}. \tag{9}$$

The APG entropy is

$$H(W) = -\sum_i w_i \log w_i. \tag{10}$$

For the binary Euclidean APG weight vector $W = (w_a, w_b)$, this gives

$$H(W) = -(w_a \log w_a + w_b \log w_b), \qquad 0 \le H(W) \le \log 2. \tag{11}$$

Remark 2.1 (Scope)**.** *The paper proves boundary dominance for the canonical APG boundary package. It does not prove the general entropy-boundary realization property for all $A \geq 1$. Instead, it proves that this realization property is exactly the final missing input required to transform APG XII into the conductor-growth theorem used by APG XIII.*

3 Review of the APG IX–XIII Route

The five-paper route toward (1) is as follows. APG IX proves source concentration:

$$\|F_{\mathrm{APG}}\|^2_{L^2(X)} \le C_1 H(W)^2. \tag{12}$$

APG X combines APG VIII and APG IX to obtain spectral entropy control:

$$E_{\mathrm{APG}} \le C_2 \lambda_1(\Delta_{\mathrm{APG}})^{-1} H(W)^2. \tag{13}$$

APG XI fixes the canonical metric (8) and proves

$$\lambda_1(\Delta_{\mathrm{APG}}) \ge \frac{c}{(\log N)^A}. \tag{14}$$

APG XII must connect the resulting analytic scale $H(W)^2(\log N)^A$ to $\deg(\pi) \log N$. APG XIII then combines the estimates.

The formal implication is simple. From (12) and (14),

$$\lambda_1(\Delta_{\mathrm{APG}})^{-1} \|F_{\mathrm{APG}}\|^2_{L^2(X)} \le C_1 c^{-1} H(W)^2 (\log N)^A. \tag{15}$$

Thus (1) follows if (4) holds. The present paper analyzes precisely how (4) can be obtained from conductor and modular-degree boundary growth.

4 The Canonical APG Boundary Package

APG VII identifies three boundary mechanisms that obstruct the APG–Arakelov projection theorem: cusp widths, vertical components, and Green-energy distortion. APG XII packages these terms into a single conductor-measurable quantity.

Definition 4.1 (Canonical APG boundary package)**.** *The canonical APG boundary package is*

$$\mathcal{B}_{\mathrm{APG}}(N, \pi) = B_{\mathrm{cusp}}(N, \pi) + B_{\mathrm{vert}}(N, \pi) + B_{\mathrm{Green}}(N, \pi). \tag{16}$$

4.1 Cuspidal boundary complexity

Let $\mathcal{C}(N)$ be the set of cusps of $X_0(N)^*$. Let $w(c)$ be the width of a cusp $c \in \mathcal{C}(N)$, and let $a_c \geq 0$ be the canonical APG cusp coefficient.

Definition 4.2 (Cuspidal boundary term)**.**

$$B_{\mathrm{cusp}}(N, \pi) = \sum_{c \in \mathcal{C}(N)} a_c \log w(c). \tag{17}$$

The normalization is chosen so that the total APG mass assigned to the cuspidal boundary is bounded by modular degree:

$$\sum_{c \in \mathcal{C}(N)} a_c \leq C_0 \deg(\pi). \tag{18}$$

This is the natural APG XII normalization because π measures the arithmetic complexity with which $X_0(N)^*$ maps into E.

4.2 Vertical arithmetic complexity

Let q range over primes dividing N. Let $\widehat{V}_q$ denote the vertical APG correction divisor over q.

Definition 4.3 (Vertical boundary term)**.**

$$B_{\mathrm{vert}}(N, \pi) = \sum_{q \mid N} \deg(\widehat{V}_q) \log q. \tag{19}$$

The canonical vertical normalization is

$$\deg(\widehat{V}_q) \leq C_1 \deg(\pi) v_q(N), \tag{20}$$

where $v_q(N)$ is the exponent of q in N.

4.3 Green distortion complexity

APG XI fixes the canonical metric (8). The Green distortion field is represented by ϕ_{APG}.

Definition 4.4 (Green distortion boundary term)**.**

$$B_{\mathrm{Green}}(N, \pi) = \int_X |\nabla \phi_{\mathrm{APG}}|^2_{g_{\mathrm{APG}}} \, \mathrm{d}\mu_{\mathrm{APG}}. \tag{21}$$

The canonical APG XI conductor-scale energy control is

$$\int_X |\nabla \phi_{\mathrm{APG}}|^2_{g_{\mathrm{APG}}} \, \mathrm{d}\mu_{\mathrm{APG}} \leq C_2 \deg(\pi) \log N. \tag{22}$$

This is the Green-energy distortion ceiling carried from APG XI into APG XII.

5 Conductor-Scale Boundary Dominance

Lemma 5.1 (Cuspidal conductor bound)**.** *Under the canonical cusp normalization* (18)*,*

$$B_{\text{cusp}}(N, \pi) \leq C_0 \deg(\pi) \log N. \tag{23}$$

Proof. For every cusp c of $X_0(N)^*$, the cusp width satisfies $w(c) \leq N$. Therefore

$$\log w(c) \leq \log N. \tag{24}$$

Using (17),

$$B_{\text{cusp}}(N, \pi) = \sum_{c \in \mathcal{C}(N)} a_c \log w(c) \leq (\log N) \sum_{c \in \mathcal{C}(N)} a_c. \tag{25}$$

Applying (18) gives (23). □

Lemma 5.2 (Vertical conductor bound)**.** *Under the canonical vertical normalization* (20)*,*

$$B_{\text{vert}}(N, \pi) \leq C_1 \deg(\pi) \log N. \tag{26}$$

Proof. From (19) and (20),

$$B_{\text{vert}}(N, \pi) = \sum_{q|N} \deg(\widehat{V}_q) \log q \leq C_1 \deg(\pi) \sum_{q|N} v_q(N) \log q. \tag{27}$$

Since

$$\sum_{q|N} v_q(N) \log q = \log N, \tag{28}$$

we obtain (26). □

Lemma 5.3 (Green conductor bound)**.** *For the canonical bounded Green-renormalized APG metric,*

$$B_{\text{Green}}(N, \pi) \leq C_2 \deg(\pi) \log N. \tag{29}$$

Proof. By definition (21),

$$B_{\text{Green}}(N, \pi) = \int_X |\nabla \phi_{\text{APG}}|^2_{g_{\text{APG}}} \, d\mu_{\text{APG}}. \tag{30}$$

The canonical APG XI Green-renormalized metric normalization gives (22). Hence (29) follows. □

Theorem 5.4 (Boundary Dominance under Canonical APG Normalization)**.** *There exists a constant $C > 0$ such that*

$$B_{\text{cusp}}(N, \pi) + B_{\text{vert}}(N, \pi) + B_{\text{Green}}(N, \pi) \leq C \deg(\pi) \log N. \tag{31}$$

Proof. By Lemmas 5.1, 5.2, and 5.3,

$$
\begin{aligned}
B_{\text{cusp}} + B_{\text{vert}} + B_{\text{Green}} &\le C_0 \deg(\pi) \log N + C_1 \deg(\pi) \log N + C_2 \deg(\pi) \log N && (32)\\
&= (C_0 + C_1 + C_2) \deg(\pi) \log N. && (33)
\end{aligned}
$$

Set $C = C_0 + C_1 + C_2$. ☐

6 Direct Entropy-to-Boundary Comparison in the Logarithmic Case

The next result is non-circular. It does not assume (4); instead, it proves the entropy-to-boundary comparison directly in the case $A = 1$ by using the boundedness of binary entropy and the existence of a conductor-detecting boundary contribution.

Hypothesis 6.1 (Conductor-detecting cusp)**.** *There exists a cusp $c_N \in \mathcal{C}(N)$ and a constant $a_0 > 0$ such that*

$$a_{c_N} \ge a_0, \qquad w(c_N) \ge C_N^{-1} N \tag{34}$$

for an absolute constant $C_N \ge 1$.

Theorem 6.2 (Direct APG entropy-boundary lower comparison for $A = 1$)**.** *Assume Hypothesis 6.1. Then there exists $C_4 > 0$ such that*

$$H(W)^2 \log N \le C_4 \left(B_{\text{cusp}} + B_{\text{vert}} + B_{\text{Green}}\right) \tag{35}$$

for all sufficiently large N.

Proof. For a binary APG weight vector, (11) gives

$$H(W)^2 \le (\log 2)^2. \tag{36}$$

Therefore

$$H(W)^2 \log N \le (\log 2)^2 \log N. \tag{37}$$

By Hypothesis 6.1,

$$B_{\text{cusp}} = \sum_{c \in \mathcal{C}(N)} a_c \log w(c) \ge a_{c_N} \log w(c_N) \ge a_0 \log(C_N^{-1} N). \tag{38}$$

For sufficiently large N,

$$\log(C_N^{-1} N) = \log N - \log C_N \ge \frac{1}{2} \log N. \tag{39}$$

Thus

$$B_{\text{cusp}} \ge \frac{a_0}{2} \log N. \tag{40}$$

Since $B_{\mathrm{vert}} \geq 0$ and $B_{\mathrm{Green}} \geq 0$,

$$B_{\mathrm{cusp}} + B_{\mathrm{vert}} + B_{\mathrm{Green}} \geq \frac{a_0}{2} \log N. \tag{41}$$

Equivalently,

$$\log N \leq \frac{2}{a_0} \left(B_{\mathrm{cusp}} + B_{\mathrm{vert}} + B_{\mathrm{Green}}\right). \tag{42}$$

Combining this with (37) gives

$$H(W)^2 \log N \leq \frac{2(\log 2)^2}{a_0} \left(B_{\mathrm{cusp}} + B_{\mathrm{vert}} + B_{\mathrm{Green}}\right). \tag{43}$$

Set $C_4 = 2(\log 2)^2/a_0$. □

Corollary 6.3 (APG XII dominance in the logarithmic spectral case)**.** *If the APG XI spectral exponent satisfies $A = 1$ and Hypothesis 6.1 holds, then*

$$H(W)^2 \log N \leq C_5 \deg(\pi) \log N. \tag{44}$$

Proof. The left side is bounded by the APG boundary package by Theorem 6.2. The boundary package is bounded above by $C \deg(\pi) \log N$ by Theorem 5.4. Combining the two estimates proves (44). □

7 The General Spectral Entropy Boundary Realization Problem

For a general spectral-gap exponent $A \geq 1$, the direct proof above is not enough. The boundary package must detect not merely $\log N$ but the full APG spectral entropy scale $(\log N)^A$.

Problem 7.1 (Spectral Entropy Boundary Realization Problem)**.** *Prove that there exists a constant $C_4 > 0$ such that*

$$H(W)^2 (\log N)^A \leq C_4 \mathcal{B}_{\mathrm{APG}}(N, \pi) \tag{45}$$

for the canonical APG boundary package.

This problem is not circular: it is a lower comparison from entropy-spectral complexity to concrete arithmetic boundary data. It is stronger than the boundary upper estimate (31). If (45) is established, then the APG XII milestone follows immediately.

Theorem 7.2 (Conditional conductor–modular degree dominance)**.** *Assume the Spectral Entropy Boundary Realization Property (45). Then*

$$H(W)^2 (\log N)^A \leq C_3 \deg(\pi) \log N. \tag{46}$$

Proof. By (45),

$$H(W)^2 (\log N)^A \leq C_4 \mathcal{B}_{\mathrm{APG}}(N, \pi). \tag{47}$$

By Theorem 5.4,

$$\mathcal{B}_{\mathrm{APG}}(N, \pi) \leq C \deg(\pi) \log N. \tag{48}$$

Therefore

$$H(W)^2(\log N)^A \le C_4 C \deg(\pi)\log N. \tag{49}$$

Set $C_3 = C_4 C$. $\square$

8 Consequences for APG XIII

APG XIII is a synthesis theorem once APG XII supplies (46). Combining (12), (14), and (46) gives

$$\lambda_1(\Delta_{\mathrm{APG}})^{-1}\|F_{\mathrm{APG}}\|^2_{L^2(X)} \le C_1 c^{-1} H(W)^2(\log N)^A \tag{50}$$

$$\le C_1 c^{-1} C_3 \deg(\pi)\log N. \tag{51}$$

Thus

$$\lambda_1(\Delta_{\mathrm{APG}})^{-1}\left\|\rho_{\mathrm{APG}} - \frac{1}{\mathrm{Vol}(X)}\right\|^2_{L^2(X)} \le C\deg(\pi)\log N. \tag{52}$$

This is exactly Open Problem 15.1 in the APG 0 formulation, provided the APG XII realization property is proved for the general exponent A.

9 Numerical Sanity Checks

This section gives scale-level checks rather than proof. For a binary APG weight vector, $H(W)^2 \le (\log 2)^2 \approx 0.48045$. If $A = 1$, then the analytic scale is at most

$$0.48045\log N. \tag{53}$$

A conductor-detecting cusp contributes at least $c\log N$ to B_{cusp}. Hence the logarithmic case is naturally consistent with direct boundary realization.

For $A > 1$, the scale is

$$H(W)^2(\log N)^A, \tag{54}$$

which cannot be controlled by a merely logarithmic cusp contribution unless additional vertical or Green distortion contributions detect the higher logarithmic power. This explains why Problem 7.1 is the correct remaining APG XII problem.

10 Conclusion

APG XII establishes the conductor-scale upper dominance of the canonical APG boundary package:

$$B_{\mathrm{cusp}} + B_{\mathrm{vert}} + B_{\mathrm{Green}} \le C\deg(\pi)\log N. \tag{55}$$

It also proves a direct non-circular entropy-to-boundary comparison in the logarithmic spectral case $A = 1$. For general $A \ge 1$, it isolates the Spectral Entropy Boundary Realization Problem:

$$H(W)^2(\log N)^A \le C_4\mathcal{B}_{\mathrm{APG}}(N,\pi). \tag{56}$$

If this problem is solved, APG XII supplies the conductor–modular degree dominance estimate required by APG XIII. The contribution of the paper is therefore both structural and technical: it proves conductor-scale boundary control and identifies the exact final arithmetic-growth theorem needed for the APG spectral-conductor program.

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